

FLS Essay Assignment Introduction

Session 3 - Tutorial
Dr Kuan Yoow Chan



UNIVERSITY OF
BIRMINGHAM

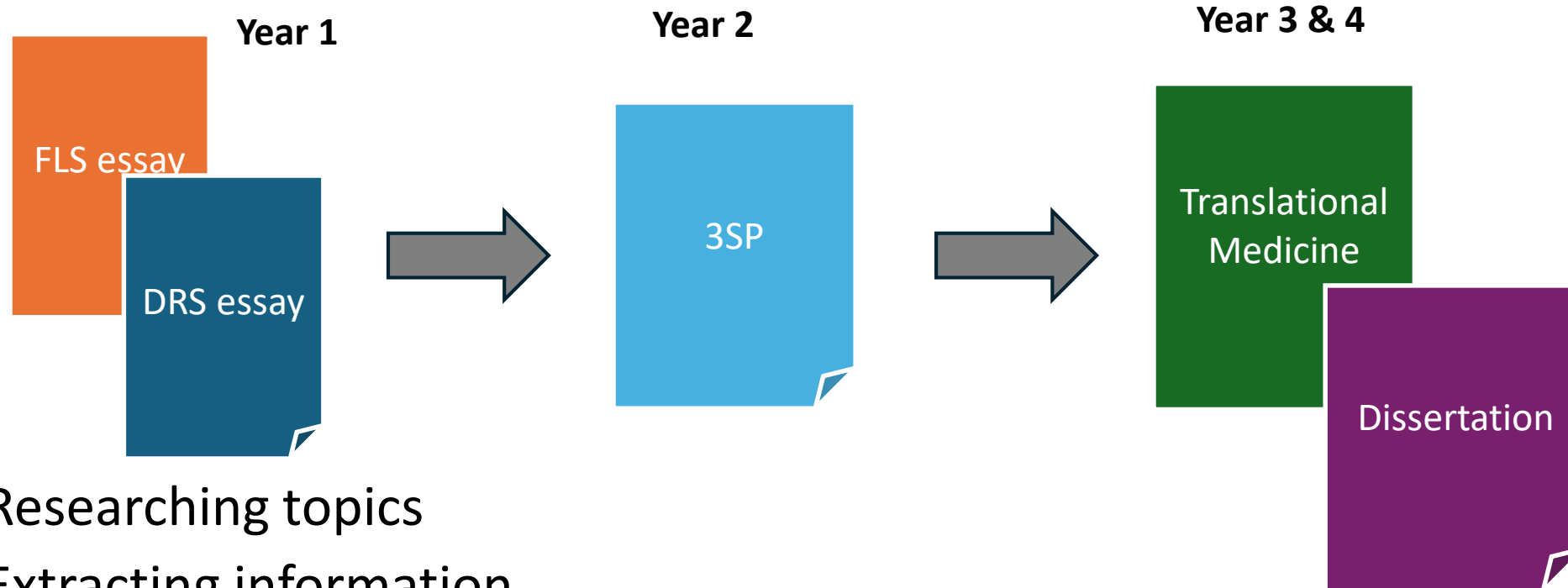
BSc Biomedical Science
Fundamentals of Life Science

Learning outcomes

- Get introduced to the essay ICA assessment
- Using an example disease (Cystic fibrosis) learn to conduct effective research on the disease
- Using an example learn how to generate an essay plan

Scientific writing

- A key skill that you will develop throughout your degree



- Researching topics
- Extracting information
- Drawing figures
- Referencing

The Task

Write an essay on the biology of a genetic disease

- **Introduce** the disease, its symptoms and consequences.
- Discuss what is known about the **gene** mutated in this disease and the **biology** of the **protein** it encodes
- Include **two specific mutations** found in the disease and their consequences on the encoded protein
- **Contrast the differences** at the molecular, cellular and body system level in normal and affected individuals
- Include **current treatments** and the development of **future therapies** focussing on how these relate to the biology of the disease
- Selection of **2 sources** you found most useful and **explanation** of why you selected them and how they helped you with the task



Key features of the essay

- The overall word count is 1500 words (+/- 150) long – i.e. between 1350-1650 words (**except figure legends or reference list**)
- Written in your own words
- Begin with a ~150-word Abstract summarising the content of your essay
- Specifically address the question
- Organised logically with sections that clearly introduce, discuss and conclude the topic
- Contain figures that are numbered and referred to from the text, have a figure legend and reference the source of the figure (if it is not your own work)
- Contain 5-10 references correctly cited to your sources of information
- Your references should include appropriate textbook(s) and/or journal review articles
- At the end you should include a 100 – 150 word reflection where you note your two most useful sources and explain why this was the case.

Essay structure

Task: Introduce the disease, its symptoms and consequences.

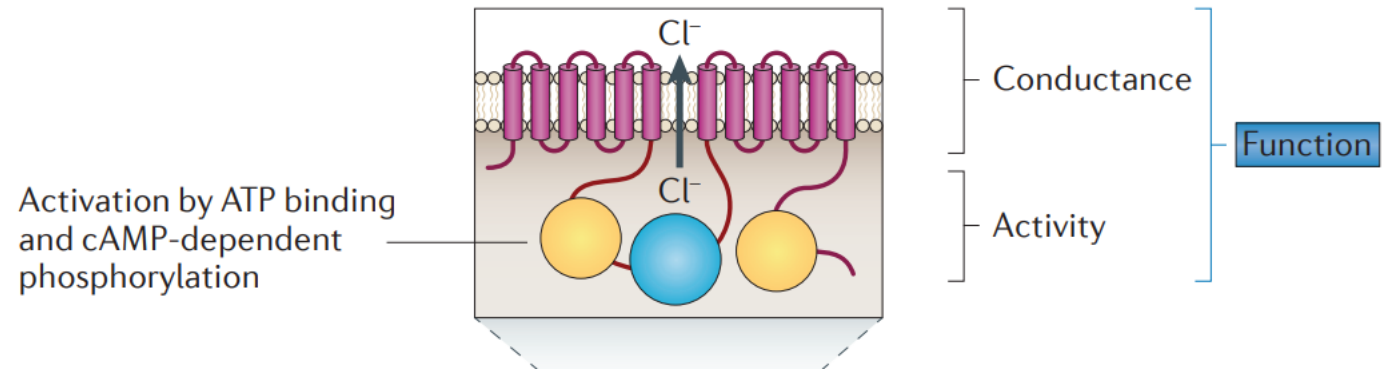
- Classification of genetic disease
- Gene involved
- Incidence
- Key Symptoms
- Prognosis

Cystic Fibrosis

- Autosomal recessive
- Mutations in the *CFTR* gene which encodes a ...
- Abnormally viscous secretions in lung and pancreas leading to...
- 1/3200 incidence in Caucasian population
- Median lifespan of 44.4

Essay structure

Task: Discuss what is known about the **gene** mutated in this disease and the **biology** of the **protein** it encodes



Twenty-five years ago, a variant (p.Phe508del; also known as F508del in legacy nomenclature) in the cystic fibrosis transmembrane conductance regulator (*CFTR*) gene was found to be the most common cause of cystic fibrosis^{3,4,5}. Demonstration that CFTR functions as a chloride channel regulated by cyclic AMP (cAMP)-dependent phosphorylation⁶ was consistent with the ion transport disturbances documented in cystic fibrosis tissues^{7,8}. Key insights into cystic fibrosis pathophysiology were derived from the study of CFTR mutants⁹, correlation of CFTR dysfunction with the cellular manifestations of cystic fibrosis¹⁰, and elucidation of protein partners involved in biogenesis and membrane function¹¹. Identification of disease-causing variants in *CFTR* contributed a tool for both the

Essay structure

Task: Include **two specific mutations** found in the disease and their consequences on the encoded protein

(Class) Impact on expression and/or function		Examples
(I) Impaired protein production	(I) Impaired protein production due to NMD	G542X R553X W1282X Y122X
	(VII) Absence of full length mature mRNA	dele2,3(21kb) 1717-1G>A
(II) Impaired trafficking		F508del N1303K G85E
(III) Impaired gating		G551D S1251N

Found an article Farinha et al, with a table summarising the different types of mutations on the CFTR gene.

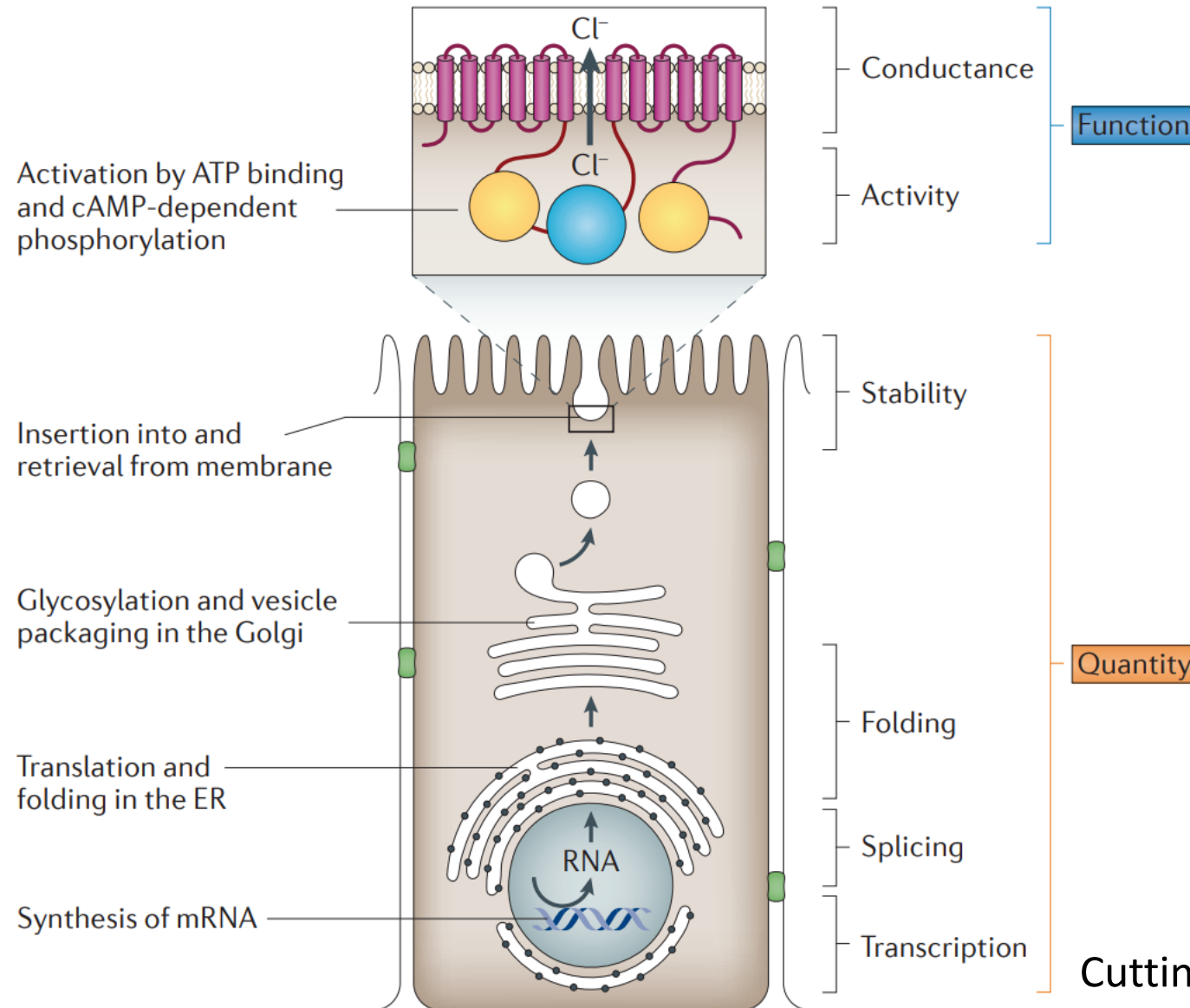
Additional useful information, include detailed information on the impact of the mutation at the molecular level

Farinha CM, Callebaut I. Molecular mechanisms of cystic fibrosis - how mutations lead to misfunction and guide therapy. Biosci Rep. 2022 Jul 29;42(7):BSR20212006.



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Essay structure: Selecting an appropriate figure



- While undertaking research, come across Cutting et al, with a figure that could be adapted for the essay
- Can be combined with Farinha et al paper to give a comprehensive overview on mutations and associate mutations with impact on molecular level

Cutting, G.R. (2015) <https://doi.org/10.1038/nrg3849>

Essay structure

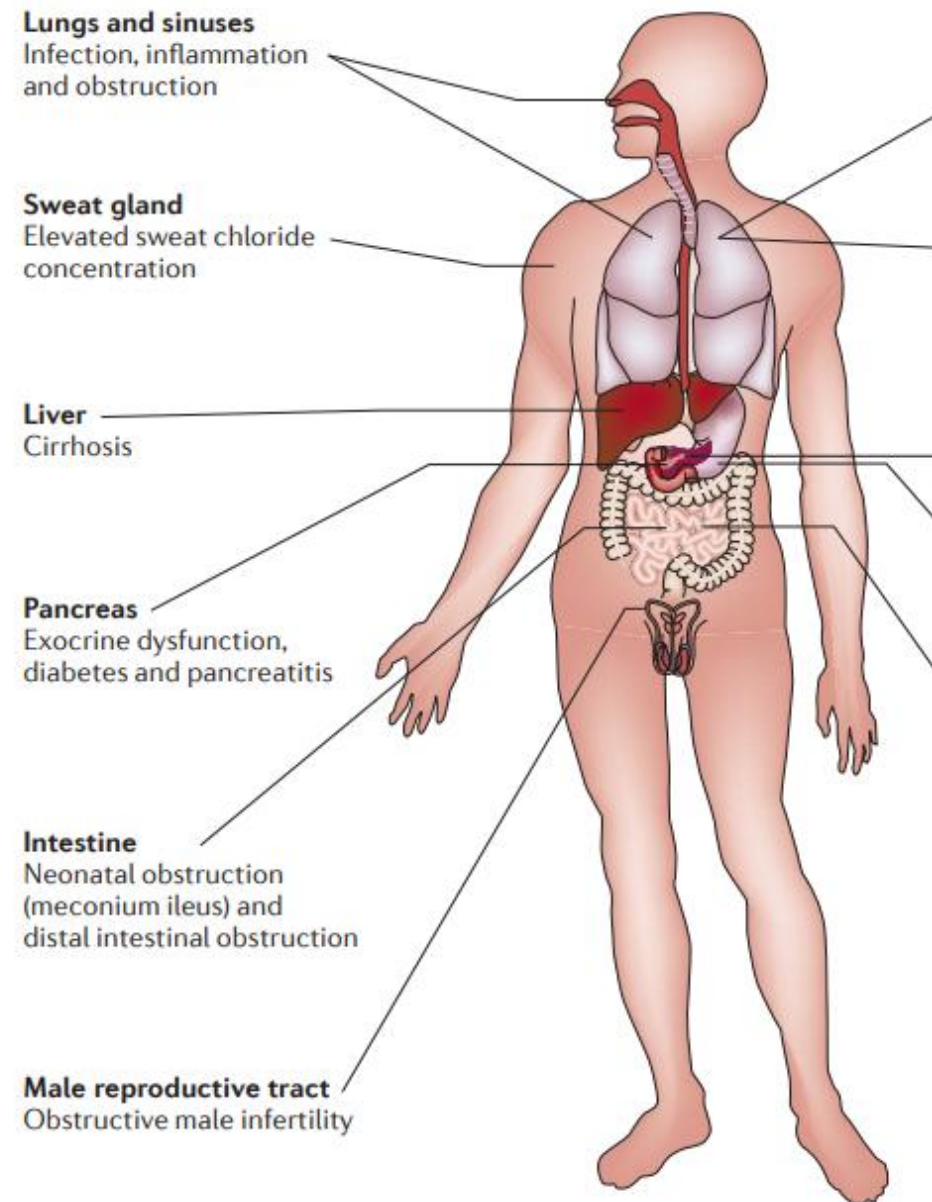
- Task: Include **two specific mutations** found in the disease and their consequences on the encoded protein
- Task: **Contrast the differences** at the molecular, cellular and body system level in normal and affected individuals
- Selected two different mutations N1303K (impaired trafficking) and G551D (impaired gating) that affect different aspects of the CFTR biology
- Examine from the aspect: How do they affect the expression of the gene, and the function of the protein produced?
 - Associate with severity
 - Associate with interactions with other proteins

Essay structure

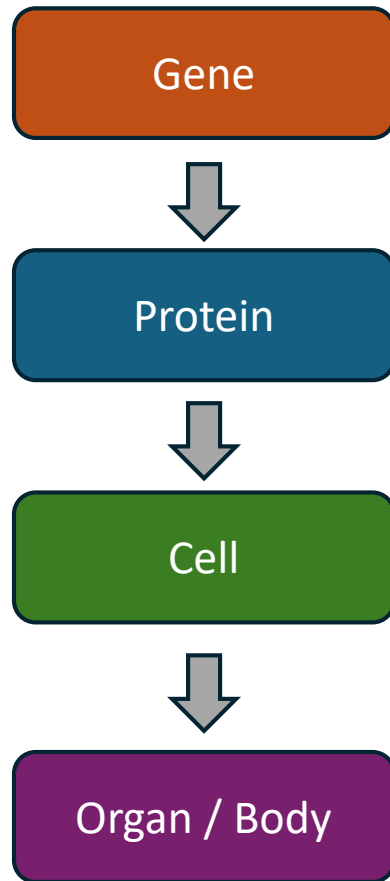
Task: **Contrast the differences** at the molecular, cellular and body system level in normal and affected individuals

Question: How do these alterations in the protein then lead to the symptoms?

Useful figure identified from Cutting et al, showing overall tissues affected



Essay structure



- What is the gene, how is it mutated?
- What protein does it encode, what does it do, how do the mutations affect its function?
- How does this altered protein affect the functioning of the cell?
- How does this affect the functioning of tissues and organs to give rise to symptoms?

Essay structure

Task: Include **current treatments** and the development of **future therapies** focussing on how these relate to the biology of the disease

- Focus on the biology – how do the treatments relate to the biology
- How might they affect the mutated protein?
- How do they alter the biology of the symptoms?

Do not give detail treatments without referring back to the biology of the disease

Table 1 CFTR mutation classes, examples, and potential pharmacotherapies^a

	Class I	Class II	Class III	Class IV	Class V	Class VI
Defect	Protein synthesis	Protein trafficking	Channel gating	Channel conductance	Reduced protein synthesis	Plasma membrane protein stability
Example mutation	G542X	F508del	G551D	R117H	3272–26A→G	c. 120del123
Therapy approach	Read-through, gene therapy	Corrector	Potentiator	Potentiator	Amplifier	Stabilizer
Available drug(s)	None	Elx/tez/iva Tez/iva Lum/iva	Iva	Iva	None	None

Use up to date references for this section

Jia, S., & Taylor-Cousar, J.L. (2023). <https://doi.org/10.1146/annurev-med-042921-021447>

^aFrom Reference 24, *Pharmacogenomics and Personalized Medicine* 2022, 15:91–104, adapted and used with permission from Dove Medical Press Ltd.

Abbreviations: elx, elexacaftor; iva, ivacaftor; lum, lumacaftor; tez, tezacaftor.

Essay structure

Task: Current treatments and **future development** of novel treatments

- What are the current challenges in treating the disease?
- What new approaches are being investigated?
- What is the underpinning biology?

Use up to date references for this section

Article found: Allen, L., Allen, L., Carr, S.B. et al. Future therapies for cystic fibrosis. Nat Commun 14, 693 (2023).
<https://doi.org/10.1038/s41467-023-36244-2>

For patients with class I (nonsense) mutations there is hope that small molecules will be identified that can facilitate premature truncation codon (PTC) read-through and/or impede mRNA decay allowing for clinically relevant levels of functional CFTR. Ribosomal read-through agents have a chequered history. Ataluren, the most extensively developed, was promising in preclinical studies and early trials but failed to demonstrate clinical benefit in larger phase three trials^{30,31}. The drug is licensed for certain variants of Duchenne muscular dystrophy, but a recent systematic review confirmed modest impacts on disease progression, but flagged the need for further work on quality of life and cost utility³². Other ribosomal read-through drugs are in preclinical or early phase clinical trial stages^{33,34}.

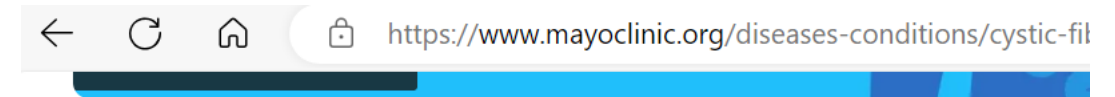
Essay structure

Abstract:

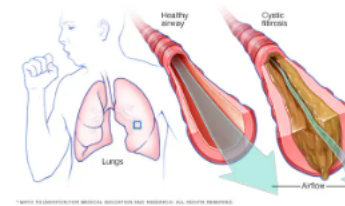
- 150 words
- Summarises key points of the essay
- Succinct (expressed clearly and in few words; concise and compressed)
- Finishes with concluding sentence giving some perspective
- Refer to essay examples for guidance
- Write last!

Researching the topic

- Some general research using the internet can be a helpful starting point
 - Patient oriented websites can give a helpful foundation – but not a suitable source



[Show transcript](#) ▼



Cystic fibrosis

[Enlarge image](#)

Cystic fibrosis (CF) is an inherited disorder that cause to the lungs, digestive system and other organs in the

Cystic fibrosis affects the cells that produce mucus, s digestive juices. These secreted fluids are normally th But in people with CF, a defective gene causes the se become sticky and thick. Instead of acting as lubricar plug up tubes, ducts and passageways, especially in tl pancreas.

Researching the topic

- Some general research using the internet can be a helpful starting point
 - Patient oriented websites can give a helpful foundation – but not a suitable source
- Generative AI can also help answer some questions you might have
 - But not a suitable source (and note what sources it uses)



Cystic fibrosis (CF) is a genetic disorder that primarily affects the transmembrane conductance regulator), which is responsible for

Key Biological Aspects of Cystic Fibrosis:

1. Genetic Basis:

- CF is inherited in an **autosomal recessive** manner, meaning develop the disease ¹ ².
- The most common mutation is the **ΔF508** mutation, which

2. CFTR Protein Function:

- The CFTR protein helps maintain the balance of salt and w
- In CF, the defective CFTR protein causes the secretions to l

3. Impact on Organs:

Learn more

1

w en.wikipedia.org

2

 med.stanford.edu

3

 britannica.com

4

 nhlbi.nih.gov

5

 mayoclinic.org

Researching the topic

- Databases such as PubMed will help you find some good review articles

- It will organise outputs by relevance
- You can filter for reviews, by date etc.
- These will be suitable sources to cite

<https://pubmed.ncbi.nlm.nih.gov/>

NIH National Library of Medicine
National Center for Biotechnology Information

PubMed®

Cystic fibrosis

Advanced Create alert Create RSS User Guide

Save Email Send to Sort by: Best match Display options

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RESULTS BY YEAR

1964 2024

TEXT AVAILABILITY

☐ Abstract

☒ Free full text

☐ Full text

ARTICLE ATTRIBUTE

☐ Associated data

ARTICLE TYPE

☐ Books and Documents

☐ Clinical Trial

☐ Meta-Analysis

☐ Randomized Controlled Trial

☒ Review

☐ Systematic Review

PUBLICATION DATE

☐ 1 year

☐ 5 years

☐ 10 years

4,092 results

Page 1 of 410

Filters applied: Free full text, Review. [Clear all](#)

☐ **Cystic fibrosis: current therapeutic targets and future approaches.**

1 Rafeeq MM, Murad HAS.

Cite J Transl Med. 2017 Apr 27;15(1):84. doi: 10.1186/s12967-017-1193-9.

PMID: 28449677 [Free PMC article.](#) [Review.](#)

Share OBJECTIVES: Study of currently approved drugs and exploration of future clinical development pipeline therapeutics for **cystic fibrosis**, and possible limitations in their use. METHODS: Extensive literature search using individual and a combination of key words relate ...

☐ **Cystic Fibrosis Modulator Therapies.**

2 Jia S, Taylor-Cousar JL.

Cite Annu Rev Med. 2023 Jan 27;74:413-426. doi: 10.1146/annurev-med-042921-021447. Epub 2022 Aug 16.

PMID: 35973718 [Free article.](#) [Review.](#)

Share **Cystic fibrosis** (CF) is an inherited multisystemic disease that can cause progressive bronchiectasis, pancreatic endocrine and exocrine insufficiency, distal intestinal obstruction syndrome, liver dysfunction, and other disorders. ...However, the advent of highly ef ...

☐ **Cystic Fibrosis.**

3 Dickinson KM, Collaco JM.

Cite Pediatr Rev. 2021 Feb;42(2):55-67. doi: 10.1542/pir.2019-0212.

PMID: 33526571 [Free PMC article.](#) [Review.](#)

Share **Cystic fibrosis** (CF) is one of the most commonly diagnosed genetic disorders. Clinical characteristics include progressive obstructive lung disease, sinusitis, exocrine pancreatic insufficiency leading to malabsorption and malnutrition, liver and pancreatic dysfunct ...

☐ **Clinical Phenotypes of Cystic Fibrosis Carriers.**

4 Polgreen PM, Comellas AP.

Cite Annu Rev Med. 2022 Jan 27;73:563-574. doi: 10.1146/annurev-med-042120-020148.

PMID: 35084992 [Free PMC article.](#) [Review.](#)

Share **Cystic fibrosis** (CF) is an autosomal recessive genetic disorder caused by mutations in CFTR, the **cystic fibrosis** transmembrane conductance regulator gene. ...

Researching the topic

- Databases such as Pubmed will help you find some good review articles
 - It will organise outputs by relevance
 - You can filter for reviews, by date etc.
 - These will be suitable sources to cite
- <https://pubmed.ncbi.nlm.nih.gov/>
- Findit@Bham
 - Wide range of sources
 - Accessible (online or in print form)
 - Can filter by type of source

The screenshot displays the FindIt@Bham search interface. At the top, the University of Birmingham logo is on the left, and navigation links for 'NEW SEARCH', 'DATABASE A-Z', 'BROWSE BY', and 'INTERLIBRARY LOAN REQUEST' are on the right. The search bar contains 'cystic fibrosis' with a search icon and a dropdown menu set to 'Everywhere'. Below the search bar, there are icons for 'All', 'Print Books', 'eBooks', 'Articles', 'Databases', 'Archives', and 'Resource Lists'. A yellow banner encourages signing in for complete results and item requests. The main results area shows 'Tweak my results' options, including 'Expand beyond Library Resources' (disabled), 'Sort by Relevance', and filters for 'Availability' (39 available in library, 108,725 full text online, 86,136 peer-reviewed journals, open access) and 'Resource Type' (141 eBooks). The search results list two items: 1. An eBook titled 'Cystic fibrosis / Dr Alex Horsley, Dr Steve Cunliffe' by Alex Horsley, published by Oxford University Press in 2015. 2. A book titled 'Cystic fibrosis / guest editor, Laurie A. Whittam' published by Saunders in 2007. Both items have a 'View Online...' link.

Create an essay plan

- A series of informative subheadings in which you will include your information
- The first subheading should be 'Introduction' and the final one is 'Conclusion'
- Include 3-5 bullet points of key information that you are planning to include in each section
- Decide on which figures you intend to use and include a brief description of the figure under the relevant subheading
- Do not include any figures or your sources in your plan – your tutor will not be able to comment on these.

Feedback opportunity: Submit essay plan and 200 words of the essay (not the abstract) by **10am on Monday 13th April 2026**

Example essay plan for Cystic Fibrosis (Part 1)

Essay Plan: The Molecular Biology of Cystic Fibrosis

1. Introduction

Hook: Begin with a brief overview of Cystic Fibrosis (CF) as a common, life-shortening genetic disease.

Context: Introduce it as an autosomal recessive disorder caused by mutations in the CFTR gene, leading to abnormally viscous secretions, particularly in the lungs and pancreas.

Key Statistics: State the incidence (approx. 1/3200 in Caucasians) and current median lifespan (~44.4 years) (Reference: Dickinson & Collaco, 2021).

Thesis Statement: This essay will explore the molecular basis of CF by examining the CFTR gene, the impact of specific mutations on the CFTR protein's biology, and how this knowledge directly informs the development of current and future targeted therapies.

Example essay plan for Cystic Fibrosis (Part 2)

2. The CFTR Gene and Protein Biology

The Gene: Identify the gene involved: the Cystic Fibrosis Transmembrane Conductance Regulator (CFTR) gene.

The Protein: Describe the normal function of the CFTR protein—an ion channel located on the surface of epithelial cells, responsible for transporting chloride ions across cell membranes. This process is crucial for maintaining the fluidity of mucus.

Classification of Mutations: Introduce the concept that different mutations affect the CFTR protein in distinct ways (e.g., protein production, processing, gating, or stability). This framework, as detailed in reviews like Cutting (2015) and Farinha & Callebaut (2022), is critical for understanding disease variability and treatment.

Example essay plan for Cystic Fibrosis (Part 3)

3. Case Study: Two Specific Mutations and Their Molecular Consequences

Focus: Contrast two mutations that disrupt the CFTR biology at different points—N1303K and G551D.

N1303K (Impaired Trafficking): This mutation causes a misfolding of the CFTR protein.

Cellular Consequence: The misfolded protein is recognized by cellular quality control mechanisms and is degraded, preventing it from ever reaching the cell surface. This results in a near-complete loss of functional protein at the apical membrane.

G551D (Impaired Gating): This mutation allows the CFTR protein to be processed and trafficked to the cell surface normally.

Cellular Consequence: However, the channel's "gate" is defective. It fails to open properly in response to cellular signals (e.g., ATP binding), severely reducing the flow of chloride ions despite the protein being present in the correct location.

Severity: Briefly note that both mutations lead to severe CF symptoms due to loss of channel function, but by entirely different molecular mechanisms.

Example essay plan for Cystic Fibrosis (Part 4)

4. From Cellular Dysfunction to Systemic Symptoms

The Biological Cascade: Explain how the defective ion transport at the cellular level leads to the characteristic symptoms at the organ/body system level.

Cell: Reduced chloride secretion (and often hyperabsorption of sodium) leads to dehydration of the airway surface liquid.

Tissue/Organ: This results in the formation of abnormally viscous mucus in the lungs and pancreas.

Body System: The thick mucus obstructs airways (leading to chronic infections and inflammation) and pancreatic ducts (impairing digestion). (Reference: Cutting, 2015, for figure illustrating affected tissues).

5. Current and Future Therapies: Targeting the Underlying Biology

Emphasize that treatments are designed to correct the specific molecular defects.

Targeting Gating Mutations (e.g., G551D): Introduce Potentiators (e.g., Ivacaftor / Kalydeco). These are drugs that bind to the CFTR protein at the cell surface and force the defective gate to remain open longer, restoring ion flow.

Targeting Trafficking Mutations (e.g., N1303K): Introduce Correctors (e.g., Lumacaftor, Tezacaftor, Elexacaftor). These act as molecular chaperones, helping the misfolded CFTR protein to fold correctly, escape degradation, and traffic to the cell surface.

Current Challenges: Discuss that not all mutations are responsive to current modulators (e.g., nonsense or splice mutations). Combination therapies (like Trikafta) are a major advance, but challenges remain in treating rare mutations and managing long-term disease.

New Approaches: Briefly mention research into next-generation correctors, mutation-agnostic therapies, and genetic approaches (e.g., gene therapy) that target the root cause at the DNA or RNA level, referencing the "underpinning biology" as the guide for these innovations. (Use up-to-date references for this section).

Example essay plan for Cystic Fibrosis (Part 6)

6. Conclusion

Summarize the direct link between the mutated CFTR gene, the altered protein's biology, and the clinical symptoms of CF.

Reinforce how a deep understanding of these molecular mechanisms has been pivotal in the rational design of highly effective modulator therapies.

Conclude with a forward-looking statement on how ongoing research into the biology of CF promises to address current therapeutic gaps and further improve patient outcomes.

Guidance on writing the essay (part 1)

- Use precise and concise writing: present accurate and complete information.
- Make sure that you understand the factual and conceptual content of your essay
- Do not use one sentence paragraphs, sentences containing all information on a certain topic should be grouped into a single well-structured paragraph
- Sections should have informative titles – these should briefly, but explicitly describe the contents of that section

Guidance on writing the essay (part 2)

- Abstract: Provide a clear and succinct summary of what is in your essay - write it after you have completed your essay.
- The essay should start with an Introduction section that introduces the disease, its symptoms and consequences
- After the introduction start with the first section that you put in your plan, make sure there is a logical flow to your different sections.
- Your final section should be a conclusion: Briefly in 3-4 sentences reiterate the key points providing insight on current understanding and future management and treatment of the disease. Do not introduce any new content into this section.

Look back through this list when you have finished writing your essay – have you followed the guidance?

Past essay examples with feedback

Essay examples
available on FLS
Canvas “Module
Assessment
Overview” page

Key dates:

Guidance on FLS essay assessment: Week 3

Deadline for submission of essay plan: 13 April 2026 (10 am)

Personalised feedback to be returned on essay plan: 27 April 2026

Deadline for submission of essay: 18 May 2026 (10 am)

Essay ICA guidance can be found [here](#)

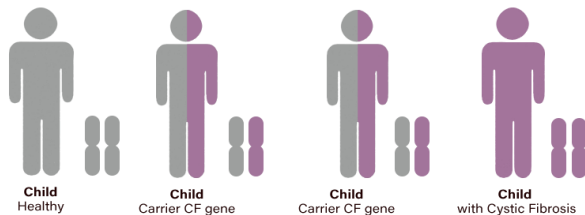
Marking criteria can be found [here](#)

Past examples

FUNDAMENTALS OF LIFE SCIENCES

Grading	Essays
1st (>70%)	1 FLS Essay - low 1st + comments.pdf ↓ ⚠
2.1 (65-69%)	3 FLS Essay 2.1 + comments.pdf ↓ ⚠
2.2 (60-64%)	5 FLS Essay - 2.2 - comments.pdf ↓ ⚠

- 



Correctors
(lumacaftor)

Production
correctors
(ataluren)

Guidance on figures

- Use 2 – 3 figures in your essay
- Number them in the order they appear
- Refer to them from the text
- You can use them directly from articles, with a citation.
- You can adapt them, also with a citation
- You can draw your own – Biorender (<https://www.biorender.com/>) is a good platform for this.

DNA is comprised of antiparallel DNA molecules (Figure 1).

Figure 1: **Structure of DNA.** DNA is made of two strands each of which can act as a template for the formation of a new strand (Travers & Mushkhelishvili, 2015).

Figure 1: **Structure of DNA.** DNA is made of two strands each of which can act as a template for the formation of a new strand (adapted from Travers & Mushkhelishvili, 2015).



Guidance on referencing

- To acknowledge an intellectual debt to another author where you have made mention of their published work
- To support specific facts or claims which you make in your text
- To enable a reader to find sources to which you have referred quickly and easily

Guidance on referencing

- Use Harvard referencing – APA 7th style
- References are cited in the text using the authors name(s) in brackets and date of publication in the text
- If there are three or more authors the surname is followed by ‘et al.’ meaning ‘and others’
- Sources are listed alphabetically in the reference list
- Quote a reference after the sentence(s) in which it has been used
- One source is sufficient for one piece of information, but if you use two (or more) different sources for information in the same sentence then they both must be quoted

Cystic fibrosis is an autosomal recessive disease resulting from mutations in the *CFTR* gene (Cutting, 2015). In recent years drugs have been developed that are targeted toward specific *CFTR* mutations found in cystic fibrosis (Jia & Taylor-Cousar, 2023).

References

Cutting, G. R. (2015). Cystic fibrosis genetics: from molecular understanding to clinical application. *Nat Rev Genet*, 16(1), 45-56.

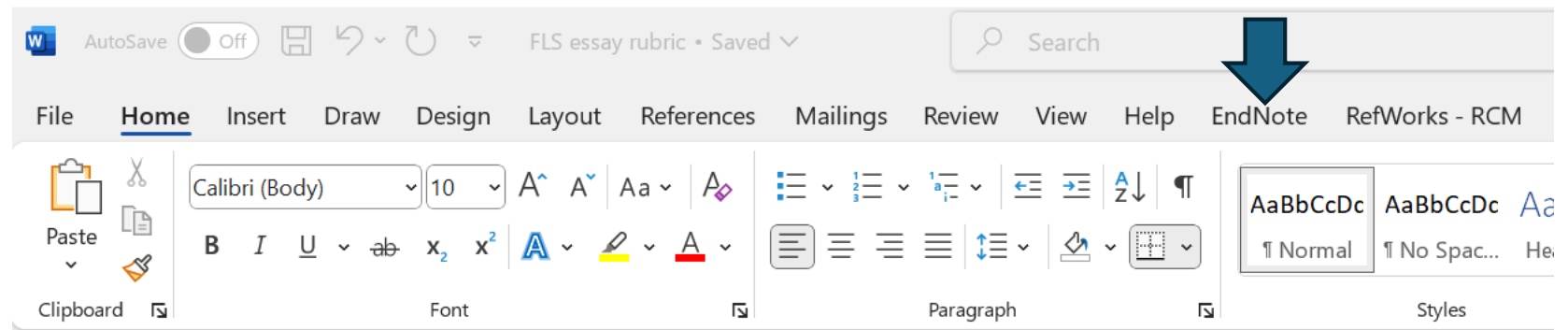
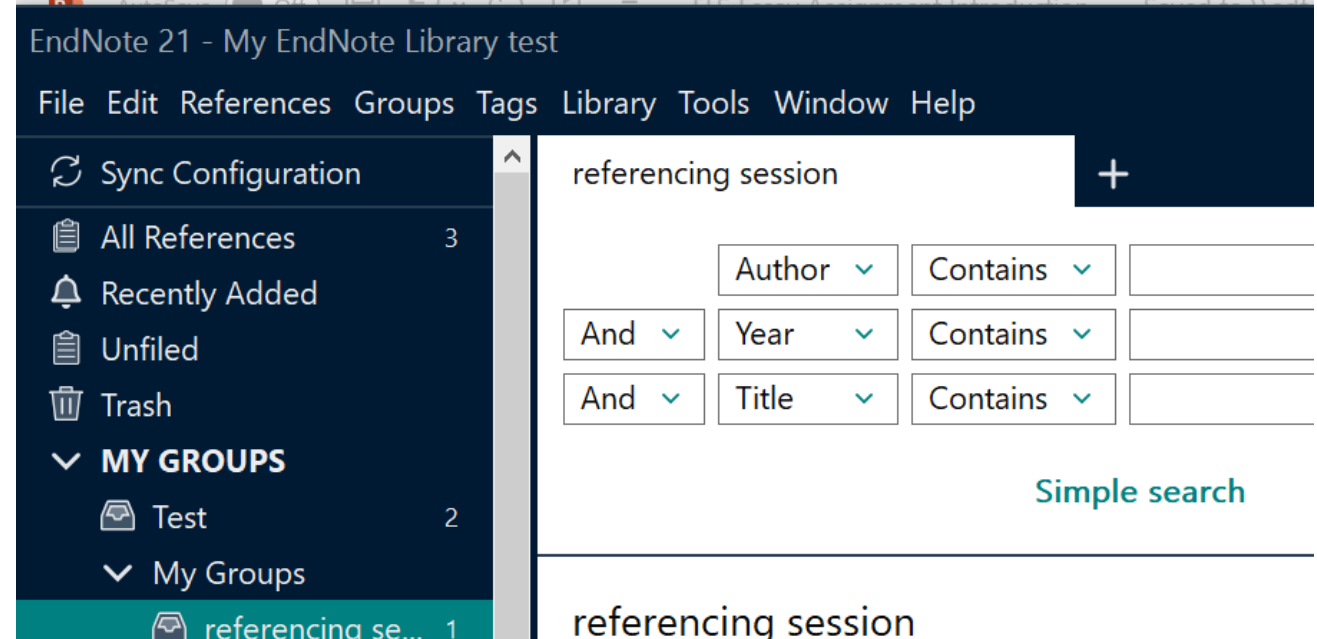
<https://doi.org/10.1038/nrg3849>

Jia, S., & Taylor-Cousar, J. L. (2023). Cystic Fibrosis Modulator Therapies. *Annu Rev Med*, 74, 413-426. <https://doi.org/10.1146/annurev-med-042921-021447>

Guidance on referencing

- Endnote in conjunction with Word will enable you to reference easily
- The software will organise and format your reference list
- Available for all UoB students:

<https://intranet.birmingham.ac.uk/student/libraries/research/endnote/index.aspx>



Reflection on your two most useful sources

- Select your two most useful sources
- Include a 100 – 150 word reflection on why they were useful and how they helped you with this assignment
- No right or wrong answer but good engagement with the task is included in the marking rubric

It might be the contents...

- Great overview of the topic
- Gives a good starting point for your research by linking to other articles
- Excellent figures
- Clear explanations that helped you understand the content

It might be how its written...

- Clear and concise writing style that is easy to follow and understand
- Structure and flow
- Good example of how to use sources
- Etc..

Note – this does not feature in the examples on the bank of assessed work

Generative AI

Acceptable Use

- 
- Researching topics
 - Understanding concepts
 - Time management and planning the task

It is not permissible to use it do the following:



- The Code of Practice stipulates that you cannot use the output of Generative AI (i.e., the content it creates) in any assessment

Detailed guidance on the use of generative AI:

<https://intranet.birmingham.ac.uk/student/libraries/asc/student-guidance-gai.aspx>

Rubric and Marking Scheme

- There are five different criteria that will be assessed
 - Abstract (10%)
 - Content (30%)
 - Quality of Sources (including the reflection on the sources) (20%)
 - Scientific writing (20%)
 - Presentation (20%)
- Refer to detailed marking criteria
- An opportunity to look at an example of an essay and how it would be graded in the assessment tutorial

Submission

- Files must be saved as either .doc or .docx or .pdf
- You must use the following format for your file name:
 - Your student i.d._Essay_FLS.doc
 - g. 1234567_Essay_FLS.doc
- Your ID number should be in the header of your essay
- Do **NOT** put your name anywhere on the document or in your file name. It is programme policy to assess work anonymously.

Abstract

Category	Exceptional >80	Excellent 70-80	Very good 60-70	Good 50-60	Satisfactory 40-50	Fail <40
Abstract	An excellent synopsis of the essay written succinctly as a paragraph within the word limit and with excellent clarity. Covers all key points of the essay and finishes with an excellent concluding sentence.	An excellent summary of the essay, contains the majority of key points and suitable concluding sentence. The writing could be more succinct in places. Written as one paragraph and within the word limit.	A paragraph summarising the essay within the word limit. Contains minor lapses in clarity in places but covers most main points of the essay. May lack a suitable concluding sentence.	A paragraph that contains some of the main points but lacking some key points. It may be too long or short and lacking clarity in places. Lack of concluding sentence. May be inappropriately formatted <u>eg</u> includes citations or abbreviations not qualified.	An attempt at an abstract with one or two relevant points. Poorly written – could be confusing or repetitive. Multiple feature of inappropriate formatting <u>eg</u> – multiple paragraphs, bullet points. May have unqualified abbreviations, citations etc.	Missing abstract, or paragraph that does not resemble an abstract in any way.

Detail

Relevance – addressing question
Include examples of mutations

Content

Understanding

Accuracy

Use of figures

Category	Exceptional >80	Excellent 70-80	Very good 60-70	Good 50-60	Satisfactory 40-50	Fail <40
Content 30%	Appropriate and relevant content, writing in such a way to include considerable detail within the word count. Demonstrates an excellent understanding of the topic area with clear demonstration of knowledge of the underlying biology in all aspects of the essay. Contains current information. Inclusion of excellent figures exemplifying key points clearly. The final concluding paragraph gives an excellent perspective on the subject area. Includes two examples of mutations occurring in the disease and a thorough description of their impact on the encoded protein and its function.	Appropriate and relevant content well-structured with a logical flow, containing key information and reference to the underlying biology in all aspects of the essay. A very good understanding demonstrated with occasional minor lapses. Includes two examples of mutations occurring in the disease and a clear description of their impact on the encoded protein and its function. Two to three figures which support the content very well-generated or taken from a suitable source. The final concluding paragraph gives a good sense of perspective.	<u>Very good content</u> , has made a good attempt to explain the underlying biology but there is some detail lacking or some inaccuracies and good understanding of the content displayed in most of the essay. Some aspects of the essay may not make good reference to the biology of the disease (eg the treatments). Includes two examples of mutations occurring in the disease but the description of how they impact the protein may lack some precision / clarity. Two or three useful figures – they make lack some detail or contain detail not relevant or explained in the main text. Inclusion of a concluding paragraph that gives some points of perspective, does not repeat large amounts of the main content	Mostly relevant content but which lacks sufficient depth and detail in a number of areas. There are some lapses in accuracy and places where good understanding has not been demonstrated. The figures may include some relevant information but do not support the text well. The essay may not be logically structured or maybe excessively descriptive with poor reference to the biology of the disease. A very poor or missing conclusion – may repeat a large amount of content. There may be either one or no specific example of a mutation, or the description may be incomplete.	Some relevant content but written at a very superficial level giving little evidence of good understanding of the disease. Some major inaccuracies or key omissions. No figures, or figures that do not support the content well.	Lacking relevant information – little to no understanding evident from the the text and figures.

Scientific writing

Succinct style

Terminology

Paragraph structure

Abbreviations

Spelling and Grammar

Category	Exceptional >80	Excellent 70-80	Very good 60-70	Good 50-60	Satisfactory 40-50	Fail <40
writing	style, using the correct scientific terminology with high levels of clarity and fluency. Excellent sentence structure, spelling and grammar. Very good use of paragraphs, no single sentence paragraphs. All abbreviations qualified. No contractions (eg don't, isn't)	scientific writing style – clearly and succinctly written with scientific terminology, with the occasional longer or less clearly written sentence. Overall excellent spelling and grammar, but with the occasional error. Good use of paragraphs, no single sentence paragraphs. All abbreviations are qualified. No contractions (eg don't, isn't)	Generally good sentence structure and the majority of the essay is clearly written. A good use of paragraphs largely avoiding the use of single sentence paragraphs. Scientific terminology has mostly used with lapses in places. Generally, grammar and spelling are very good but with the occasional error. The majority of abbreviations are qualified and the use of contractions largely avoided.	that the content can be understood. The scientific writing style may be poor in places or sentences may be poorly constructed such that they are repetitive or long and complicated. There may be use of multiple short 1-2 sentence paragraphs. Abbreviations may not be qualified and there may be considerable use of contractions. More frequent errors in spelling or grammar.	weak. Information is conveyed but there are frequent poorly constructed sent and large number of spelling and grammar errors. No/little evidence of the use of scientific terminology. Significant problems with paragraph structure, abbreviations and the use of contractions	sentences, language significantly impedes meaning / understanding. Very poor grammar, spelling and paragraph structure.

Quality of Sources

High quality sources –
review articles and text
books

Good engagement with
the reflection

Relevance

5 – 10 (no more!)

Category	Exceptional >80	Excellent 70-80	Very good 60-70	Good 50-60	Satisfactory 40-50	Fail <40
Quality of Sources 20%	Identified and used high quality sources: review articles and <u>text books</u>. All sources were up to date and highly relevant. No use of sources such as patient-oriented websites, Wikipedia etc. All the information is in the sources cited. Inclusion of a short reflection giving clear and cogent explanations of why the two sources selected from their reference list were the most useful to them.	Identified and used high quality sources: <u>text books</u> and review articles. The majority are up to date and relevant. No use of sources such as patient-oriented websites, Wikipedia etc. All the information is in the sources cited. Inclusion of a short reflection giving clear and cogent explanations of why the two sources selected from their reference list were the most useful to them.	Mostly used high quality sources – the majority are review articles and <u>text books</u>. There may be one or two less useful sources (eg. patient-oriented websites) and/or less current sources. All the information is in the sources cited. Inclusion of a short reflection giving clear and cogent explanations of why the two sources selected from their reference list were the most useful to them.	Used two-three high quality sources but also two or more less suitable sources such as patient-oriented websites. May have used very outdated articles used to source information for current / future treatment. May find that some of the information included is not in the sources cited. The reflection may be missing or lacking a very clear explanation of why the selected references were useful.	<u>The majority of</u> sources are not suitable, although there are 1-2 useful sources (current <u>text books</u> or review articles). Information included in the essay is not found in the sources. Omission of a reflection or a poor explanation of why the selected references <u>were</u> useful.	No <u>high quality</u> sources No reflection.

Presentation

Organisation into subsections with informative titles: eg
Cellular function of CFTR

Presentation	Figures are appropriately sized such that any text the contain is clearly legible. Legends are formatted exactly as per the instructions and do not contain excessive descriptions. All figures are referred to from the text. Each section has a title which clearly articulates the contents of that session. Title are formatted such that they are distinct from the remainder of the text (no use of underlining) The formatting of the reference is list is excellent, formatted exactly as specifications instruct. No missing information and consistent formatting of the different elements of reference (eg <u>year</u>, journal, title etc.)	Most elements are excellent (see >80). However, there may be occasional minor lapses. For example, figures may be too small, or there may be some small inconsistencies in the formatting of the reference list.	Mostly well formatted. Figures are mostly appropriately sized and figure legend contain the majority of the relevant information but there may be some omissions. Sections have subheadings but some may be less informative. The reference list contains the majority of the key information, some minor omissions for some of the sources or the reference list has inconsistencies in the formatting.	Generally good formatting but some more significant problems. For example, figures may contain illegible text and/or legends missing key information. They may not be referred to from the text. The essays <u>is</u> lacking subsection or has too many and the subtitles may not be distinctly formatted. The reference list has key information missing for the sources or is not well formatted with poor consistency throughout.	Poor presentation. Subsections or figure legends are missing. The reference list is included but has multiple omissions and serious formatting errors.	Very poor formatting – multiple omissions such that the submission contains no or extremely few features that have been outlined in the instructions.
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Formatting of reference list:
use referencing software!

Professional formatting eg distinct font style for subheadings, no underlining

Timeline

If there are extenuating circumstances affecting your ability to complete this assignment, please email the JEP Office biomed-hust@contacts.bham.ac.uk

- **Week 3, 16th March:** Essay introduction and guidance
- **Monday 13th April 10:00 am:** Essay plan and 200 word sample submission via canvas
- **Monday 27th April:** Feedback on essay plan to be returned
- **Monday 18th May 10:00 am:** Deadline for submission
- **Friday 19st June:** Marks and feedback released
- **Please use FLS discussion board to ask any questions you may have.**

The title of the FLS essay is:

The Biology of Pompe Disease

Summary for Part 1

- Using an example disease (Cystic fibrosis) you learnt how to identify useful research sources on the disease
- Using an example disease you learnt how an essay plan is generated
- Using the marking criteria, you identified key elements of the essay that will be assessed

Part 2

Abstract writing practice



UNIVERSITY OF
BIRMINGHAM

Learning outcomes

- Learn to identify important elements of a good abstract
- Using an example identify practices that should be avoided when writing an abstract
- Using a short two page essay, practice writing an abstract



Abstract writing guidance

- 150 words
- Summarises key points of the essay
- Succinct (expressed clearly and in few words; concise and compressed)
- Finishes with concluding sentence giving some perspective
- Refer to essay examples for guidance
- Write last!

Evaluating a good vs bad abstract

- Read the following abstracts
- Based on what you have learnt about a good abstract, comment on the abstract and what can be done to improve

Abstract writing (Good example)

Ocean acidification, driven by anthropogenic carbon dioxide absorption, poses a significant threat to marine ecosystems and associated economies. This is exemplified by the Chesapeake Bay's declining shellfish populations and biodiversity. This paper examines the etiology of acidification and evaluates the efficacy and feasibility of proposed mitigation strategies. The biogeochemical process by which elevated atmospheric CO₂ increases oceanic acidity is detailed, including its direct physiological impacts on calcifying organisms such as oysters and corals, and its neurological effects on fish. Current intervention methods are critically assessed. Iron fertilization, intended to stimulate CO₂-absorbing phytoplankton blooms, is associated with adverse effects including nutrient depletion and the creation of hypoxic "dead zones," while its long-term carbon sequestration potential remains unverified. Conversely, the cultivation of macroalgae, particularly kelp forests, offers a localized, ecologically benign approach to carbon drawdown, nutrient remediation, and habitat provision. It is concluded that no singular, globally scalable solution currently exists; however, an integrated strategy combining regional interventions like kelp reforestation may offer the most viable pathway for mitigating localized acidification while broader technological breakthroughs are pursued.

Abstract writing (Good example)

Ocean acidification, driven by anthropogenic carbon dioxide absorption, poses a significant threat to marine ecosystems and associated economies. This is exemplified by the Chesapeake Bay's declining shellfish populations and biodiversity. This paper examines the etiology of acidification and evaluates the efficacy and feasibility of proposed mitigation strategies. The biogeochemical process by which elevated atmospheric CO₂ increases oceanic acidity is detailed, including its direct physiological impacts on calcifying organisms such as oysters and corals, and its neurological effects on fish. Current intervention methods are critically assessed. Iron fertilization, intended to stimulate CO₂-absorbing phytoplankton blooms, is associated with adverse effects including nutrient depletion and the creation of hypoxic "dead zones," while its long-term carbon sequestration potential remains unverified. Conversely, the cultivation of macroalgae, particularly kelp forests, offers a localized, ecologically benign approach to carbon drawdown, nutrient remediation, and habitat provision. It is concluded that no singular, globally scalable solution currently exists; however, an integrated strategy combining regional interventions like kelp reforestation may offer the most viable pathway for mitigating localized acidification while broader technological breakthroughs are pursued.

Starts with a broad statement of the problem and uses relevant examples to establish importance.

Provides the aim of the article.

Provides concise but detailed scientific overview of the problem with relevant examples

Provide some critical evaluation of current solutions by giving a balanced viewpoint

Ends with a clear take away message which is balanced

Overall, the reader is able to understand the entire message of the article: the problem, the mechanisms, the debate between solutions, the final argument without reading the main article. This is the **primary** goal of any abstract.

Good elements of a scientific abstract

- First section should highlight the topic area and why the topic is important
- Second section should discuss about the content of the essay in a concise manner with relevant examples
- Provide some concluding statements on the topic being discussed
- Language used should be formal and use accurate scientific terminology
- The abstract should be self-contained and able to provide a good overview of the article summarised

Abstract writing (bad example)

- Read the following abstract
- Point out the problems with the abstract

Abstract writing (poor example)

Ocean acidification is when the ocean gets more acidic because of CO₂. It's a big problem for the Chesapeake Bay. The Bay has lots of wildlife and ocean acidification kills oysters and fish which is bad because oysters filter the water and make it clean. This hurts the economy too because fishermen can't make money.

One plan to address this issue is to introduce iron in the ocean to make phytoplankton grow. However, this might not work because copepods eat the phytoplankton. Another idea is to plant kelp which grows really fast like 18 inches per day. Kelp helps absorb CO₂ and gives fish a place to live. The Puget Sound Restoration Fund studies this and thinks it works pretty good locally.

The essay also talks about coral reefs dying because acidification. This is really bad for biodiversity. The UN says it will cost a trillion dollars by 2100 which is a ton of money. We need a breakthrough to fix this but there's no perfect solution yet.

Abstract writing (poor example)

Ocean acidification is when the ocean gets more acidic because of CO₂. It's a big problem for the Chesapeake Bay. The Bay has lots of wildlife and ocean acidification kills oysters and fish which is bad because oysters filter the water and make it clean. This hurts the economy too because fishermen can't make money.

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The essay also talks about coral reefs dying because acidification. This is really bad for biodiversity. The UN says it will cost a trillion dollars by 2100 which is a ton of money. We need a breakthrough to fix this but there's no perfect solution yet.

- The abstract lacks the standard abstract format. It jumps between observations, economic impacts, solution evaluation, and back to personal sentiment.
- The writing style was not formal. Scientific writing should be with a passive writing style.
- While some explanation is given, it appears superficial without relevant detail to support the statements.
- Lack of use of accurate scientific terms.

Practice writing an abstract for an essay

- Download an example essay “A Plantastic Solution to an Aqueous Problem” by John Lin from the FLS Canvas page (Session 3 – Tutorial – Essay setting session).
- Read the essay and attempt to write a 150 word abstract summarising the essay
- Suggestion:
 - Read the entire essay first before deciding the key message of the essay
 - Then extract key points from the essay
 - Then compose the abstract summarising the essay with your own words

Example model abstract from the practice essay

The global water crisis, exacerbated by desertification and population growth, demands sustainable solutions beyond conventional storage methods that are labor-intensive and often unsanitary. Cactus adaptations offers a viable alternative for water harvesting and storage in arid regions. Cacti employ specialized mechanisms for water acquisition and retention, including shallow, extensive root systems that utilize capillary action, spines that condense atmospheric fog, and mucilage with natural sediment-binding properties. Additionally, waxy cuticles and protective structures minimize evaporative loss while maintaining optimal internal temperatures. By emulating these biological strategies, location-specific containers can be engineered using biodegradable materials to passively collect, filter, and preserve water. Such innovations address critical shortcomings of current methods, including the tradeoff between temperature regulation and sanitation. Without intervention, water scarcity threatens to precipitate public health crises, agricultural failure, and geopolitical conflict. Cactus-inspired biomimicry therefore represents a cost-effective, environmentally sustainable approach to securing safe water access for vulnerable populations in developing nations.

Things to remember about abstract writing

- The example abstract is one version. There are many other possible versions.
- Focus on the overall picture!
- Start with an introductory statement.
- Introduce a problem.
- Provide solutions and discuss with appropriate examples.
- Give a take home message at the end.

Summary for Part 2

- Identify the key elements of an effective abstract.
- Recognise common mistakes to avoid when writing an abstract.
- Practice writing an abstract based on a short essay.

